

Introduction to the Unit

This unit is about the land and people of Asia. The interaction between people in different Asian countries is discussed. In addition, you will learn about the physical features of land, water resources and the process of migration and settlement patterns of the people in the Asia.

CHAPTER :1 THE LAND OF ASIA**SLOs:**

- Locate Asia on a map of the world.
- Locate and name the major physical features of Asia on a map.
- Identify physical features of Asia using a topographic map.
- Examine contour lines of selected places in Asia and find the highest and lowest points.
- Identify major forests of Asia on a map.
- Explain where the rainforests of Asia are located why they are growing there.
- Describe the importance of rainforests.
- Compare past and present aerial photographs and satellite images of rainforests in Asia to show deforestation.
- Conduct an inquiry on the measures taken to preserve rainforests in Malaysia and Brazil.
- Name the major deserts of Asia.
- Describe the features of the deserts of Asia.
- Describe how deserts in some places are growing due to natural occurrences.
- Compare the characteristics of the Gobi Desert and the desert of the Arabian Peninsula.
- Describe vegetation and animal life in the deserts.
- Compare the nature of coastlines of Britain and Bangladesh.

Introduction

In this chapter, you will learn about Asia the largest continent on the Earth. Asia has variety of physical features that includes high mountains, high plateaus, fertile plains and vast deserts. It has many coastal and inland (far from sea) areas. The geographic variation affects climatic conditions of area. Different climatic conditions help variety of vegetation to grow, where many species of animals live.

Do you know?

Russia is the largest country and China is the most populated country of Asia, as well as of the world.

Location

You have learnt in Grade VI that Asia is the world's largest continent. It has the largest land size (17,212,000 square miles) and population (more than 60%) of world. The continent shares its border with Europe and the Urals (mountains) to the West.



Asia on the World

Source: www.google.com.pk

Activity

Look at the above map and name the oceans that form the borders of Asia. Teacher will help students to point out the directions North, South and East.

- To its North _____
- To its East _____
- To its South _____

Teacher's Note

For brain storming, teachers will ask students to recall directions i.e. East, West, North and South. Then, teachers will remind them (usually on the maps) East is at right hand side, West is at left hand side. North is between East and West upward and South is also between East and West but downward.



Physical Features of Asia

Source: www.google.com.pk

Mountains in Asia

Do you remember what you learned in the previous class about Himalayas? Recall where they are located and name the highest peak of Himalayas. Asia is covered with mountains, plateaus, plains and deserts. Some of the important mountains and their location are mentioned below.

Mountains	Location	Mountains	Location
Altai	East-Central Asia	Tien Shan	Central Asia
Ghats	Eastern and western coast of India	Ural	Northern edges of Russia to Kazakhstan
Kunlun	Western China and Gobi desert	Zagros	Western and southwestern Iranian plateau till Strait of Hormuz

Plateau

Plateau is a higher area than the adjacent areas. It is a high altitude area above the sea level with fairly flat top surface. Its shape is similar to the shape of table. Asia's plateaus are:



Plateau

Plateau	Location
Iranian plateau	Iran, Afghanistan and Pakistan
Deccan plateau	Southern part of India
Tibetan plateau	Southern Mongolia and western China (It is known as Rooftop of the World)

Do You Remember?

You have learnt about plateaus of Pakistan in Grade VI. There are two plateaus i.e Balochistan & Potowar plateaus.

Plains

Plain is the low and flat land. In Pakistan, Punjab and Sindh cover the Indus Plain area.

Activity

Find out any two plains of Asia (other than South Asia). The key given in the map of Physical Features of Asia will help you.

- i. _____
- ii. _____

Teacher's Note

Teachers will share knowledge with students that mostly green colour is used to show plains. Now, they should focus on green areas and find out plains. Make a list of shown plains then exclude Indo-Gangetic Plain as it is located in South Asia.

Coastlines of Asia

Asia's coastline is around 62,800 km long. Bangladesh is one of the Asian countries that are located at coast. It is low-land area with rivers that flow into Bay of Bengal. It experiences heavy monsoon rainfall that causes floods every year. These floods destroy agricultural farms and affect the human lives severely.

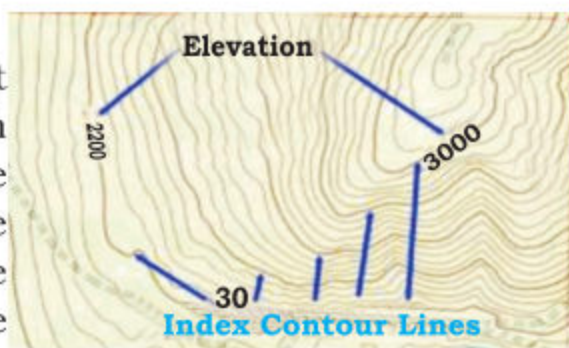
As compared to Bangladesh, Britain is a country in Europe that has many islands. It also suffers from coastal floods but not every year. Sometimes high tides during spring bring flood over here.

Do you know?

During summer, winds blow from Bay of Bengal, Indian Ocean and Arabian Sea towards land and cause Monsoon rainfall.

Topographic map

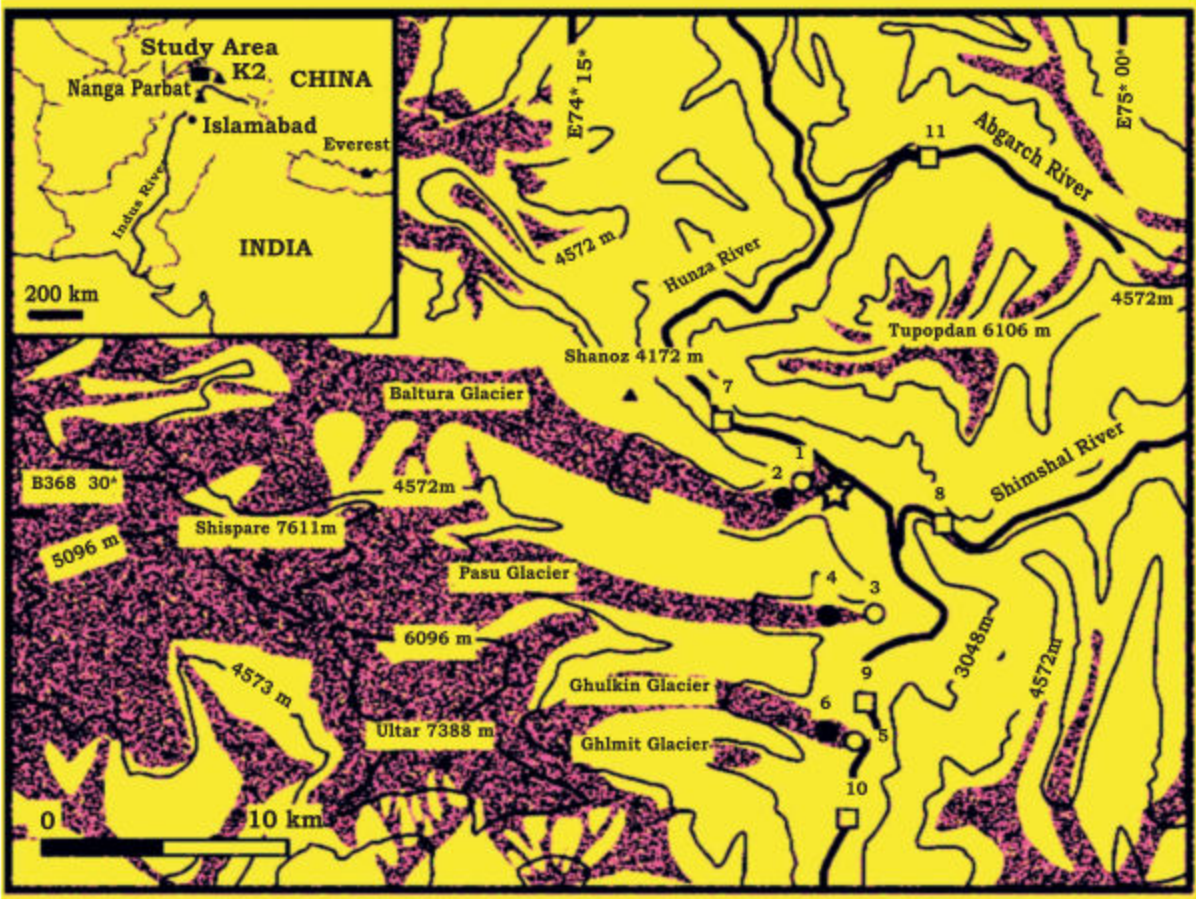
A map that shows the different elevation/ altitude in an area through lines is called as Topographic map. The elevation is shown through lines that are called as contour lines. For example, see (Contour Lines on a map) how the contour lines show elevation on the map.



Contour Lines on a map

The Highest and the Lowest Points

The surface of Earth is comprised of elevation variation. There is no large area that has same altitude and every area has the highest and the lowest point.



Karakoram Range

Activity

Study the above figure and find out the highest and the lowest points of Karakoram Range that is located in Pakistan and China.

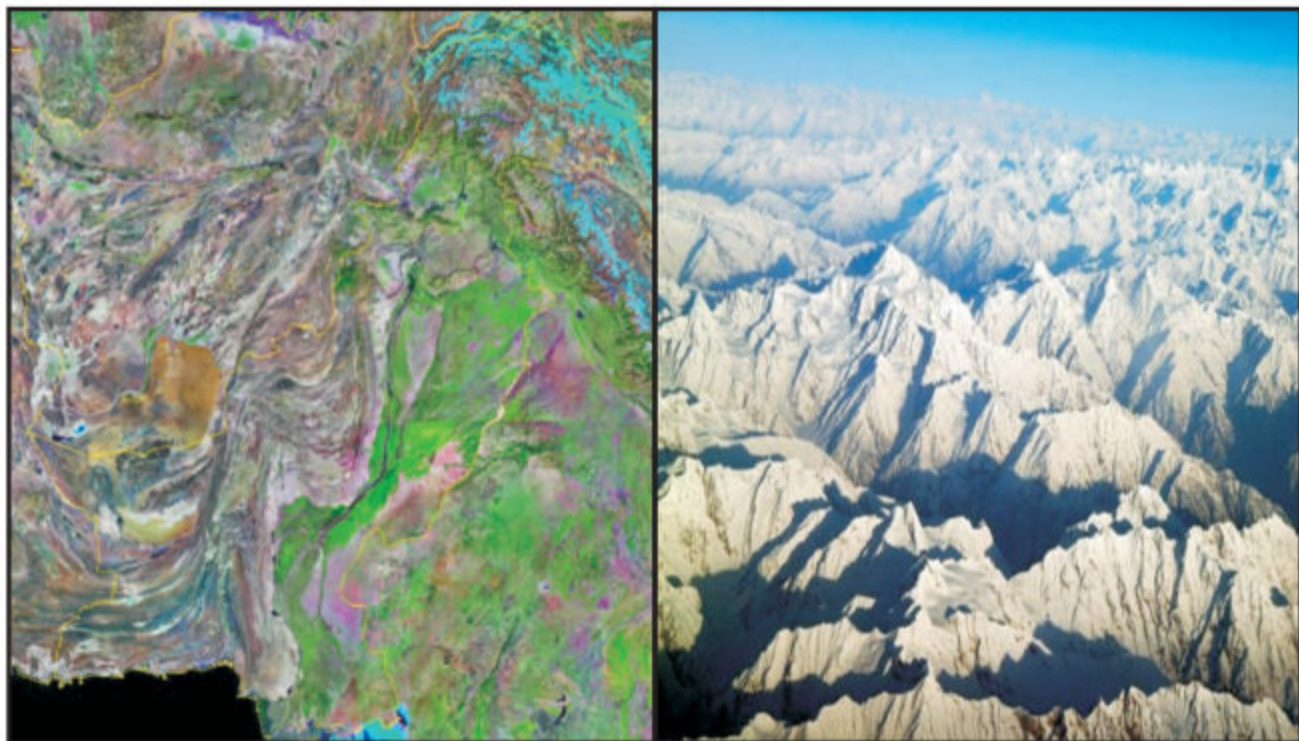
- i. The highest point _____
- ii. The lowest point _____

Teacher's Note

Teacher will take the help of degrees of adjectives e.g. good, better and best. Then, the teacher will discuss with students that the lowest point is the one that has no more lower point. Similarly, teacher will discuss the meaning of the word 'highest'. Highest is the point from where there is no more higher point.

Aerial Photograph

During flight, when we look outside the plane, buildings and land features look very small. It is called aerial view. Similarly, the photographs of ground taken from very much height is called aerial photograph. It also shows the buildings and features very small.



Aerial photograph of Karakoram Range

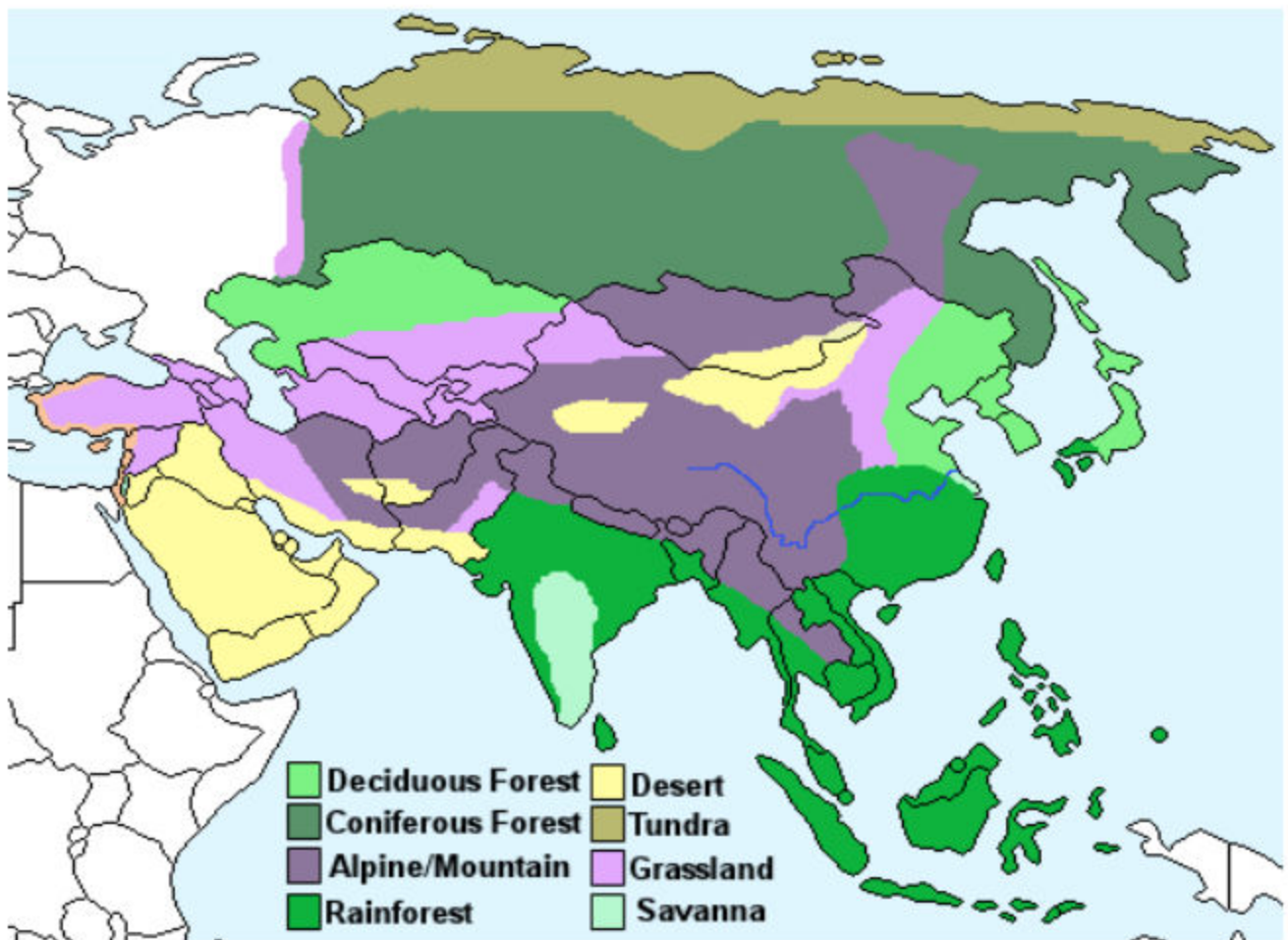
Actual Local of Karakoram Range

Activity

Compare both the photographs. You can see how they look different? Aerial photograph shows larger areas with less detail. While, actual location shows small area with more details.

Teacher's Note

Teachers can ask students about their pictures. They can discuss the effects on picture if it is captured with little or much distance.



Map of Vegetation in Asia

Source: www.google.com

Rainforests

A rain forest is a thick forest of tall trees which is found in tropical areas where there is a lot of rain. In Asia, they are grown in South and Southeast Asia. They are grown in this area which is near to equator where the Sun shines throughout the year and it experiences lots of rain. Abundance of sunlight and rain help them to grow dense.



Rainforest

Importance of Rainforests

Rainforests are very important. They supply oxygen and reduce carbon dioxide. They help stabilize the world's climate and maintain

water cycle. They protect the area from flood, drought and erosion. There are a large number of species of plants and animals which are used for medicines and foods.



Deforestation



Forestation

Rainforests are cut down everywhere. Now, Malaysia is taking measures to preserve rainforest. It arranges rallies to develop awareness among people to protect the forest. Besides, it plants the species of rare trees.

Dou you know?

A process in which a fertile land is changed into a desert is called 'desertification'.

Major Deserts of Asia

Desert is an area with annual rainfall of 10 inches or less than 10 inches. Either, it has no vegetation or little vegetation. Deserts are very hot during summer and very cold during winter.



Deserts of Asia

Source: www.google.com

Gobi Desert

Gobi Desert is located/situated in the south of Mongolia and north of China. As the area is very far from Indian Ocean, it could not receive rain from it (Indian Ocean). Thus, the desert was formed. Here winds carry desert sand and deposit the sand on fertile land that is why it is growing every year.



Gobi Desert

The following animals are found in the Gobi Desert:



Jerboa



Bactrian Camel



Golden Eagle

Arabian Desert

Arabian Desert is located in Western Asia (Middle East). The desert has little or no vegetation with hardly any human presence. The Arabian Desert was formed because Tropic of Cancer passes through the desert. When the sunlight directly falls on the land, great heat is generated. Besides it, the Arabian Peninsula is located between the Red Sea and the Persian Gulf. These small water bodies do not produce much cool winds that is why, the land remains hot. It is also the reason for scarce vegetation in the desert.



Arabian Desert

The following animals are found in the Arabian Desert:



Spiny Tailed Lizard



Caracal Kittens



Camel Spider

Thar Desert

Thar Desert is also known as The Great Indian Desert and it is located/ situated/ lies in two countries i.e India and Pakistan. Its more than two third part is located in India. From Rajasthan and Punjab in India, it extends its border into Pakistan in the provinces of Punjab and Sindh. It has been formed because Monsoon that blows from Indian Ocean towards land, bypass this area or does not enter into the area. Besides Aravalli Hills is located in parallel of wind direction that does not help/ prevents winds to lift up and complete the process of condensation. It results in no or little rain. The following animals are found in the Thar Desert:



Thar Desert (Rajasthan)



Saw Scale Viper



Great Indian Bustard



Black Buck

Karakum Desert

Karakum means black sand and the desert is named as dark coloured soil is found/ lies beneath the sandy desert. It is located east of Caspian Sea in Turkmenistan that is part of Central Asia. It is a hilly plain that is rich in minerals. The desert was formed when ancient course/ route of Amu Darya dried up and sand was deposited/left. Besides, sediments of nearby rivers Murgab and Tedzhen have been deposited here.



Karakum Desert

The following animals are found in the Thar Desert:



Karakul Sheep



Spider



Goitered Gazelle

Activity

On the world map below, shade the area of Asia. You can take the help from the world map given in the beginning of the chapter.



World Map

Source: www.google.com

Using an outline map of the Physical features of Asia, mark the places on which you find the above taught topographical features of Asia. You can take help from the map of physical features of Asia.

SUMMARY

In this chapter, you studied that Asia is surrounded by oceans from three sides. Ural Mountain forms western border of Asia. Tibetan Plateau is located at the border of Mongolia and China. Due to the variation of elevation and climatic conditions many types of vegetation have grown. Two deserts of Asia the Arabian and Gobi deserts have been formed in different ways and both have different desert animals too.

END OF THE CHAPTER EXERCISE

Check your knowledge and understanding

Fill in the blanks

- i. The land size of Asia is _____.
- ii. The coastline of Asia is _____ long.
- iii. Ural is the name of _____.
- iv. Arabian peninsula is situated between _____ and _____.
- v. Karakoram Range is located in _____ and _____.

Answer the following questions

- i. Enlist the mountains, plateaus and deserts of Asia.
- ii. If you were to live near coastlines which one will you prefer between Britain and Bangladesh, and Why?
- iii. Define the following terms:
 - a. Plain
 - b. Contour Lines Map
 - c. Summer Monsoon
 - d. Aerial Photograph
- iv. Gobi Desert and Arabian Desert have been formed in different ways. Explain the ways.

Conduct an Inquiry

1. With the help of internet, find out types of desert vegetation in Asia.
 - a. _____
 - b. _____
 - c. _____

Cooperate with others

Brazil is located in South America. Its rainforests are under threat. Is Brazil trying to preserve the forest? Search in newspaper or any magazine, in group of 4 students. Share your information with your classmates.

SLOs:

- Describe the physical and cultural characteristics of different regions of Asia (the Middle East, The South East Asian Region).
- Justify the importance of studying the different cultures of Asia and World.
- Describe how deserts in some places are growing and in others shrinking.
- Describe the changes people have made to the Arabian Desert.
- Highlight the causes of deforestation.
- Identify strategies to reduce deforestation.
- Explain the causes and effects of coastal flooding in Bangladesh.
- Describe the life of people in the rainforests of Malaysia and Indonesia.
- Describe the life of people in the coastal regions of Bangladesh and Japan.
- Use a variety of geographic tools such as aerial photographs and Geographic Information Systems (GIS) to compare places in the world and to analyze changes in landforms and human activity.
- Analyze relationships among people, places, and the environment by mapping information including trade patterns, governmental alliances, and immigration patterns.
- Compare physical and human characteristics of places using different maps to describe how environments in the world have changed (e.g. rainforests in Amazon Basin.)
- Classify and locate major natural (natural features of the land, vegetation, animals, and climate) and human environments (cities, farming areas) of the world.

Introduction

Asia is the highly populated continent in the world. It consists roughly one-third of the land of the entire world and contains about three-fifths of population. The two most populous countries of the world i.e. China and India, are part of this continent. These together account for nearly two-fifths of all people as Asia occupies a wide territory. It comprises a great range of physical features which highly affect the people of this region. Likewise, the people of Asia have a huge impact on land they live in.

Salient features of Asian Culture

People of Asia practice a variety of culture. As it is a highly populated region therefore; it is rich in heritage and traditions. Asia is well known for celebrations and festivals. It is a land where festivals of different cultures and religions are being celebrated. Among them, major festivals are Diwali, Holi, Baisakhi, and Eidul-Fitr, Dusshera, Onam, Christmas and many others are celebrated with great fervor all over India. Followers of different religions of the world are found in this part of the world. Towards the western part of Asia, Muslim countries like UAE, Qatar, Iraq, Iran and Dubai are located and contain features of Arab and Iranian cultures. Asia has people following the religions like Hinduism, Islam and Christianity.

Asia has fertile lands in China and India and excels in agricultural production of food, grains and rice. However, Southeastern countries of Nepal, India and Bangladesh along with many countries like the Islamic nations practice agricultural activities. Let's understand how people are living in different physical settings of Asia.

Significance of studying the diverse cultures of Asia and World

Understanding of other cultures is known as 'Intercultural understanding'. It helps people in interacting others in a respectful way. It brings a sense of peace and harmony. It develops social skills and allows people to cooperate with each other. In modern times, the world has become global village. There is an urge need for our multicultural society to have a strong sense of identity while respecting the values of others.



Activity

Prepare for a debate competition in class and suggest ways to develop peace and harmony. Understanding of other cultures can help to overcome global issues like terrorism.

Climate change and its effects on deserts

The following table will inform you about deserts:

Area occupied by deserts around the world	about one fifth of the Earth's surface.
Occurrence	Occur where rainfall is less than 50 cm/year.
Disturbances	Occasional fires or cold weather, and sudden, infrequent, but intense rains that cause flooding.
Central Asian countries facing desertification	Afghanistan, Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan and Uzbekistan.

Earth's climate is destabilizing because of the rising atmospheric absorptions of carbon dioxide. By the burning of fossil fuels, it causes deserts to advance or shrink. Our choice is a simple one: reverse these trends or risk being overwhelmed by them.

Activity

After understanding reasons for advancing deserts, what do you think is the possible reason of shrink of deserts? List them on a flip chart and paste charts in gallery. Go for gallery walk and see how many members share the same thoughts.

Teacher's Note

Incorporate input shared by pupils and emphasize on their validity with detailed explanation.

Effects of human activity on Arabian Desert

Arabian Desert is great desert region located on the edge of southwestern Asia occupying almost the entire Arabian Peninsula. It is the largest desert area on the continent—covering an area of about 900,000 square miles (2,300,000 square km). In the table below, activities of one of its habitant group are explained. This will help you to identify that how environment is being effected by humans.

Habitant	Activities
Bedouin	Everyday Activities
	1. Nomadic desert life by breeding camels, Arabian horses, and sheep.
	2. Grow date palms and other crops, usually hiring others to perform agricultural labour.
	Interaction with settled communities
	1. Religious rituals (e.g., the hajj [pilgrimage] to Mecca).
	2. Long-distance commerce.
	3. Exchange of poetry and other cultural activities.
	Modernization
	1. Availability of transportation.
	2. Discovery of petroleum.
	3. Settlement in urban areas(a few do part-time nomadism)
	4. Material used in construction of buildings.

The paramount natural resource of Arabian Desert is its underground water supply. Recent techniques have been used by the governments of Arab countries to develop water sources and to irrigate soils for farming. Desalinization (a process of removing salts and minerals from a substance) plants are built along the coasts to produce great quantities of fresh water from seawater, making the Arabian Peninsula one of the leading regions of the world employing that technology.



Activity

Observe the given picture carefully and note down how humans are effecting natural environment? Make a list of the problems caused by humans. Write your findings here:

Activity: Human impact on Forests

Read the boxes below and match the suitable title (a or b) for each box and suggest strategies to reduce deforestation.

Box 1

Forests have been providing a great deal of facility to humans. In particular, Asia which has a huge difference of altitude, climate and elevation all across its region. Humans have been damaging this dense vegetation for timber and cultivation of land. This eventually leads to problems like soil erosion, deforestation, overgrazing, pollution and natural disasters. This also makes animals living in these forests lose their habitats and gradually fall in category of extinct animals from endangered species as they are not taken care of. A healthy ecosystem will also help to prevent forest fires (also called wildfires).

Box 2

Forests enable stability in atmospheric temperature. Without clean atmosphere, it is impossible for humans to survive. To regulate the temperature, trees and soil work together. This process is called evapotranspiration. Trees help to rectify air pollutants. Trees keep the soil held on ground which stops soil erosion. It helps in storing the moisture and protecting land from heavy rainfall. Forests also slow down the flow of water preventing floods. Rivers flowing between forests bring silt with them making the land fertile and keep the soil moist, stopping evaporation which may prevent wild fires.

Titles

- a. Protection of the forests from deforestation and natural habitats from getting extinct.
- b. Understand ways through which people survive in natural disasters in the forest regions.

Life-style of people living in the rainforests of Malaysia and Indonesia

People in Sumatra	People in Borneo
Majority of the local People follow Muslim faith. One of the common practices is that all important acts are done only with right hand (for example, shaking hands, passing and receiving money or food). Because, the left hand is considered evil. People living there are hospitable towards tourists. Native people are happy to help them.	The main religions in Sarawak are Islam, Christianity and Buddhism. People here practice religious tolerance thus, people are free to follow their religions. The Borneo has several sub-groups and each group is completely different from each other. Earlier, people travelled by boat but, the recent road development is diminishing this tradition.

In an international review of scientists on the World's largest rainforest – Amazon was pointed out that humans are the cause of great disturbance in Amazon basin. Clearing of trees cause longer dry seasons, later leading to severe issues such as forest fires and droughts.

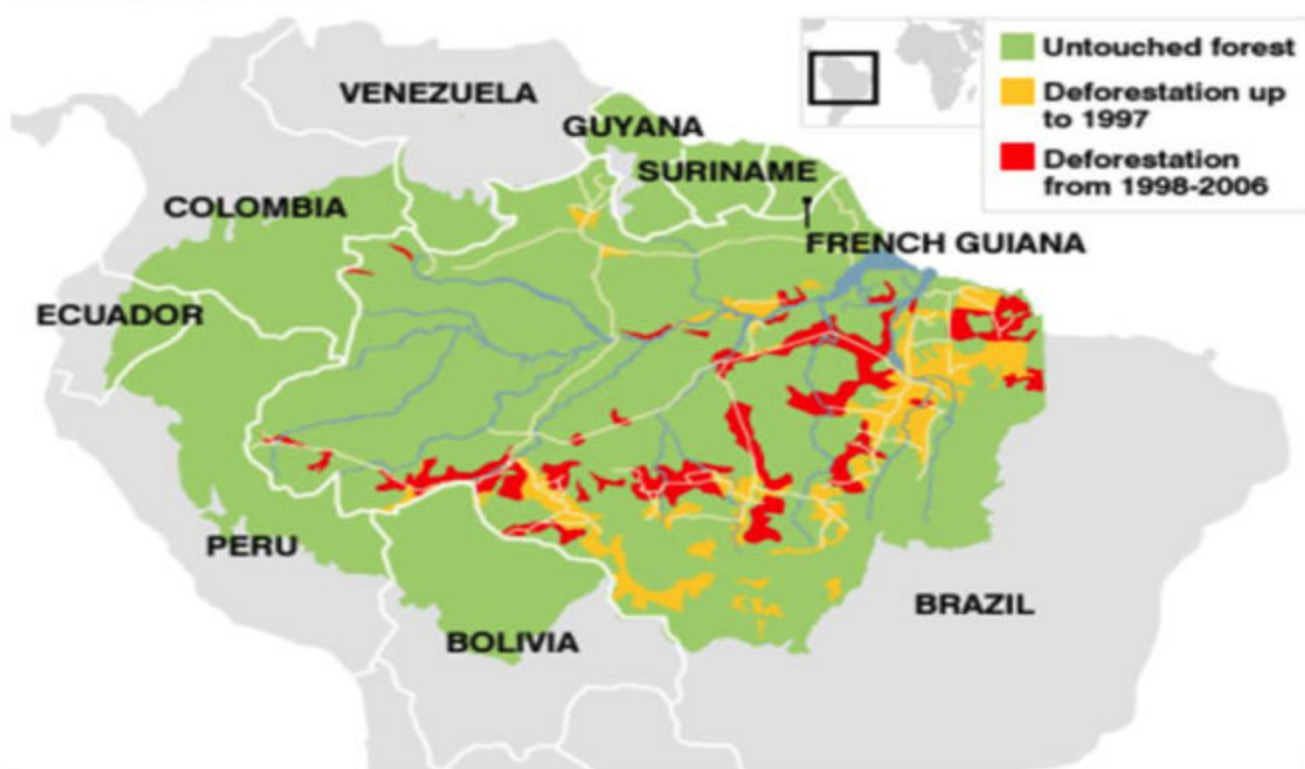
These below mentioned reasons have made a huge impact on amazon forests:

1. Agricultural expansion
2. Spread of human settlement
3. Deforestation

Do you know?

Sumatra (located in Indonesia) is the world's sixth largest island. The islands' tropical climate have created habitats that house thousands of unique species and the world's last remaining Sumatran tigers, orangutans, pygmy elephants and Sumatran rhinos. Borneo (located in Malaysia) is the world's third largest island, covering an area slightly larger than Texas.

AMAZON RAINFOREST



Activity

How Amazon is affected by human activities? Write your findings with the help of your teacher or any other adult.

Constraints of living near coastal lands

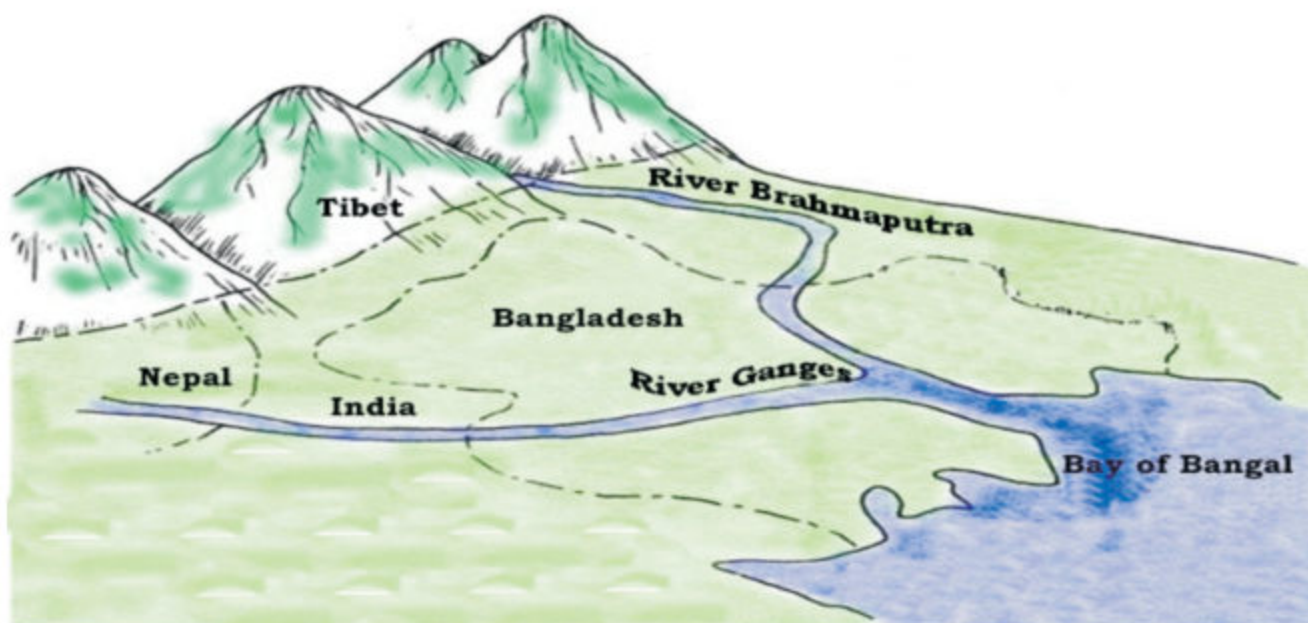
Modern constructions and recent technologies have led to a vast change in natural environment. The human activity has brought a serious threat to nature in terms of climate change. Climate change has resulted in an increase in sea level which may cause flooding or storms and high level of destruction. Scientists have claimed that a high level risk is being faced by coastal areas. At a rough estimate, more than 200 million people across world-wide live along coastlines less than 5 meters above sea level. By the end of the 21st century, this figure is estimated to increase to 400 to 500 million.

Activity

Classify physical and human causes that allow floods from the given picture and write them on black/white board after making two columns, Discuss ways of controlling human factors;

Some Causes of Flooding in Bangladesh

Monsoon climate brings heavy rain and snow. Soils are leached and heavy runoff results in soil erosion.



1. Spring Snow-Melt results in soil erosion and a rapid increase in river discharge.
2. Deforestation in Headwater Areas is due to increasing population in Nepal & Tibet. Trees cleared for fuel and grazing land. Less Evapotranspiration, more runoff and faster soil erosion. Landslides also occur.
3. Rivers silt-up due to increased soil erosion. This raises the river bed and reduces the capacity of the channel resulting in increased likelihood of flooding.
4. 80% of Bangladesh lies on a huge floodplain and delta. Most of which is only 1m above sea level.
5. Much of the Ganges has been diverted for irrigation purposes, this removes some of the silt and prevents the floodplain further downstream from being built up.
6. Cyclone (violent storms) frequently hit Bangladesh.

Activity

List four ways of controlling floods.

- 1- _____
- 2- _____
- 3- _____
- 4- _____

Activity

Read the information below and find similarities and differences between Life of people in the coastal regions of Bangladesh and Japan . Write your findings through a Venn diagram.

Life-style of residents in the coastal regions of Bangladesh

It is one the developing countries of Asia. Due to its geography, it faces cyclones affected by the climate change. The climate change has led to untimely rainfall, hotter summers and irregular monsoons. The flash floods cause damage to crops resulting in financial crisis for farmers and people living in coastal areas. The people are either homebound or migrate because of health issues (dengue, malaria, diarrhea), injuries and lack of opportunities.

Life-style of residents in the coastal regions of Japan

Japan is one of the developed countries of Asia. It is a mixture of old and new cities. It has cities with skyscrapers, subways, underground plazas etc. However, there is another side of Japan having in villages which are not completely rural. The villages which are located near urban settlements provide opportunities to commuters and industrial workers. Problems like environmental pollution, constant menace, earthquakes and floods are also faced by Japan.

SUMMARY

Asia is a highly populated continent with a wide range of cultural, political and religious identities. As world is becoming closer, it is necessary to have inter-cultural practices to maintain world peace and harmony. People are practicing a variety of life style in certain areas of this continent. Climate change is causing rise in sea level which results in floods, advancing of deserts and shrinking of deserts too. Arabian Desert is the largest desert in the continent with desalinization plants to meet the needs of water supply. Borneo and Sumatra are two best examples of tropical rainforests in Asia. However, many natural disasters occur due to deforestation for which human activity is to be blamed. Whereas, Bangladesh faces coastal flooding due to its low-lying feature. Therefore, it can be said the physical and human activities are responsible in effecting the environment and there is an urge need for us of do out bit.

END OF CHAPTER EXERCISE

Check your knowledge and understanding

Fill in the blanks

- The two most populous countries of the world i.e China and _____ are part of Asia.
- The major crop grown in Bangladesh is _____.
- Borneo is located in _____.
- The capital city of Malaysia is _____.
- _____ is a process of removing salt from substance.

Answer the following questions

- Suggest ways of protecting forests from deforestation.
- Write a note on economic activities of the people of Asia living in Arabian Desert.
- What do you know about climate change and its effects on people of Asia?
- Deserts lack water. What methods are used by local people of deserts to store water?
- If you are given a chance to live a habitat of your choice, where would you prefer to live and why? Share reasons.

Use your inquiry/investigation skills

Complete the given table and analyze relationships among people, places and the environment by mapping information including trade patterns, governmental alliances and immigration patterns.

Physical feature	Occupation	Why is it important for the local people ?	Governmental alliances	Reasons for settling in a particular place
Desert				
Forest				
Coastal land				

Cooperate with others

Imagine yourself living in a rainforest. Which one do you prefer to live in? What would be possible activities of the people living here?

In groups, make a table on a chart and classify major natural (natural features of the land, vegetation, animals, and climate) and human environments (cities, farming areas) found on that rainforest.

SLOs:

- List the sources of fresh water.
- Suggest ways in which water can be conserved by individuals and communities.
- Describe the water cycle (use the terms precipitation, condensation, evaporation, transpiration, run off through flow and ground water in the description).
- Explain how the water cycle affects river systems.
- Define the terms erosion, weathering, transportation and deposition.
- Describe the importance of rivers.
- Describe the effects of rivers on the landscape (e.g. weathering, erosion, transportation, deposition).
- Describe the upper course, middle course and lower course of a river.
- Describe the landscapes that are formed as the river runs its course (eg use the example of the Indus).
- Explain people's use of river in environments.
- Explain how people affect a river's flow.
- Describe how water can be beneficial as well as destructive.
- Explain how floods are caused.
- Identify countries in Asia that are affected by floods.
- Explain the process of rehabilitation of flood affectees in Sindh.
- Design a brochure for Sindh highlighting a strategy to minimize losses caused by floods.
- Explain why dams are built.
- Examine the social and human impact of large dams in Asia e.g. India's Narmada river and China's three Gorges dam project.
- Explain irrigation system and the major problems caused by irrigation in Pakistan.
- Describe the canal system and its types in Pakistan.
- Identify the causes of water logging and salinity.
- Highlight the measures taken by government to indicate the problems of water logging and salinity.
- Explain the functions of tube-well technology.
- Make a pamphlet for farmers on changing irrigation practices and identify benefits of these practices.
- Explain cyclones/typhoons and how they occur.
- Identify countries in Asia that are at risk from cyclones.
- Identify the causes and effects of water pollution.
- Suggest solutions to the problem of water pollution.
- Discuss the causes of drought.

Introduction

In this chapter, you will learn about fresh water, water cycle and uses of water. You will study about the flow of water and its problems such as flood, shortage of fresh water and effects of flowing water on human life and environment.

What is Fresh Water?

Freshwater is defined as water with a low concentration of salt — usually less than 1%. Water is the most common substance found on earth. It exists in all three types of matter: as a gas in the form of water vapours and steam, as liquid in the form of water and as solid in the form of ice.

Sources of Fresh Water

There are two major sources for fresh water:

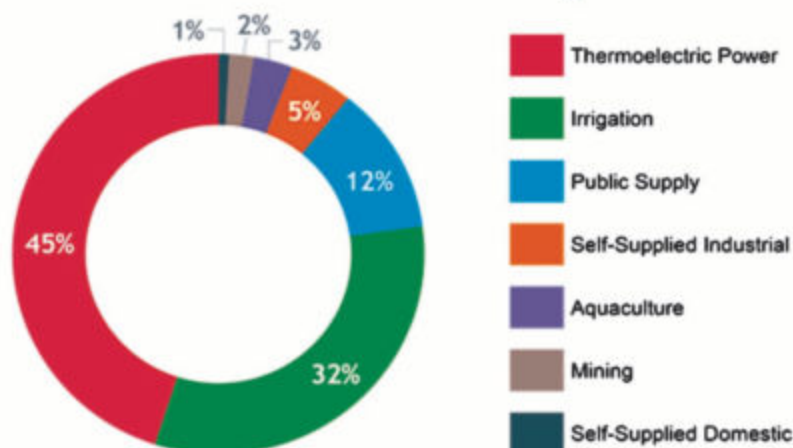
Ground Water - Water that is found in thin aquifers (layer of soil that absorbs water) under the earth's surface is known as ground water. This water is generally found around 2,000 feet deep in the Earth.

Surface Water – Water that is found on the surface of the earth is known as surface water. Water in streams, rivers, lakes, reservoirs and glaciers are the examples of surface water.

Water Consumption

Imagine a world with no water at all. Would you be able to drink it, bathe, and swim and so on? Look at following graph that shows water consumption. You can see that the amount of water is used in a number of ways such as to grow crops, to manufacture goods and to keep businesses running smoothly. We also use a significant amount of water to meet the nation's energy needs.

Water Consumption



Do You Know?

Water covers about 75 percent (%) of the earth's surface, in which 97% of water is in oceans and it is too salty for people, animals or plants to use.

- 2% water is frozen at the north and south poles in glaciers and on snowy mountain ranges.
- Only 1% is fresh water but not all of it can be used.

Why Is Conserving Water Important?

The Earth has abundant water, but in fact less than 01 percent is available for human use. Demand of fresh water resources is increasing day by day due to increasing population, while the supply of fresh water is always constant. Although, water cycle continuously returns water to the Earth, it is not always returned to the same place or in the same quantity and quality.

Do You Know?

Humans depend on natural resources and water is the most valuable natural resource. If we continue to waste and pollute water, we will destroy all lives on earth including human life.

Activity

Draw posters individually or in groups to suggest ways in which water can be conserved. Show individuals and groups of people being wise with water. Share/present your posters with school mates to help them to understand how they can contribute to conserve water.

Teachers' Note: Teachers will ask students to suggest how we as an individual can save water. For example, we should try to consume minimum water for bathing etc.

How Is Water Recycled?

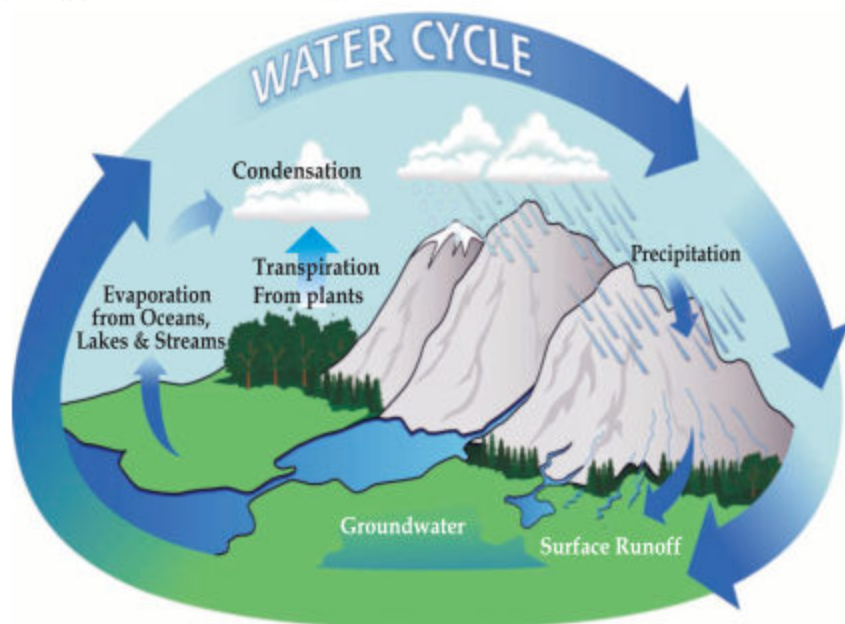
The water cycle has been working for billions of years. The water cycle describes the existence and movement of water on, in, and above the Earth. Earth's water is always in movement and is always in changing states, from liquid to vapours to ice flakes and back again.

Look at the picture (on next page); you see that water cycle starts with **evaporation**. It is a process where water at the surface turns into water vapors as gas. Water absorbs heat energy from the sun and gets changed into vapours. Water bodies such as the oceans, the seas, the lakes and rivers are the main sources of evaporation.

As water vaporizes into water vapors, it raises up in the atmosphere. At high altitudes, the water vapours change into very tiny particles of ice flakes/ water droplets, because the temperature at a high altitude is low. This process is called **condensation**. These particles come closer to each other and form clouds and fog in the sky.

Then, the water droplets combine to make bigger droplets. When air cannot hold bigger droplets any more, it precipitates. The clouds pour down as precipitation due to change in wind or temperature. As, at high

altitudes, the temperature is low, the droplets lose their heat energy. These water droplets fall down as rain. If the temperature is very low (below 0 degree), the water droplets fall as snow.



As water precipitates, some of it is absorbed by the soil and used by plants. This water enters into the process of **transpiration**. Transpiration is a process almost similar to evaporation in which water is turned into water vapours by the plants.

As the water on the soil pours down, it leads to **runoff**. Runoff is the process in which water runs over the surface of earth. When the snow melts into water, it also leads to runoff. As water runs over the ground, it displaces the surface of soil and takes away /moves the minerals to the stream. This runoff combines to form water channels and then rivers. It ends up into lakes, seas and oceans.

The water that does not flow into the rivers, it moves deep into the soil. This is called **infiltration**. The water seeps down and increases the level of ground water table. It is called pure water and is drinkable.

How Water Cycle Affects River Systems?

A river is very important part of the water cycle which carries rain water back to the sea. Rivers can run for many hundreds of miles to reach sea or ocean. Not all the water flowing down the rivers is returned to the sea or ocean. Some of the water that we use in/at our homes, schools, shops, offices, factories etc, comes from rivers. When we use water, we remove it from the natural water cycle and return it to rivers via sewage treatment systems.

Actions of Water:

Erosion



Erosion is the action of water, wind or ice that removes sediments of soil, rock, or dissolved material from one location to another.

Weathering



Weathering is breaking down or disintegration of substances such as rock and minerals by physical or chemical processes. It is the gradual destruction of rock under surface conditions, breaking it down into progressively smaller pieces.

Transportation



Transportation is defined as a movement of eroded particles of rocks by wind, water or ice.

Deposition



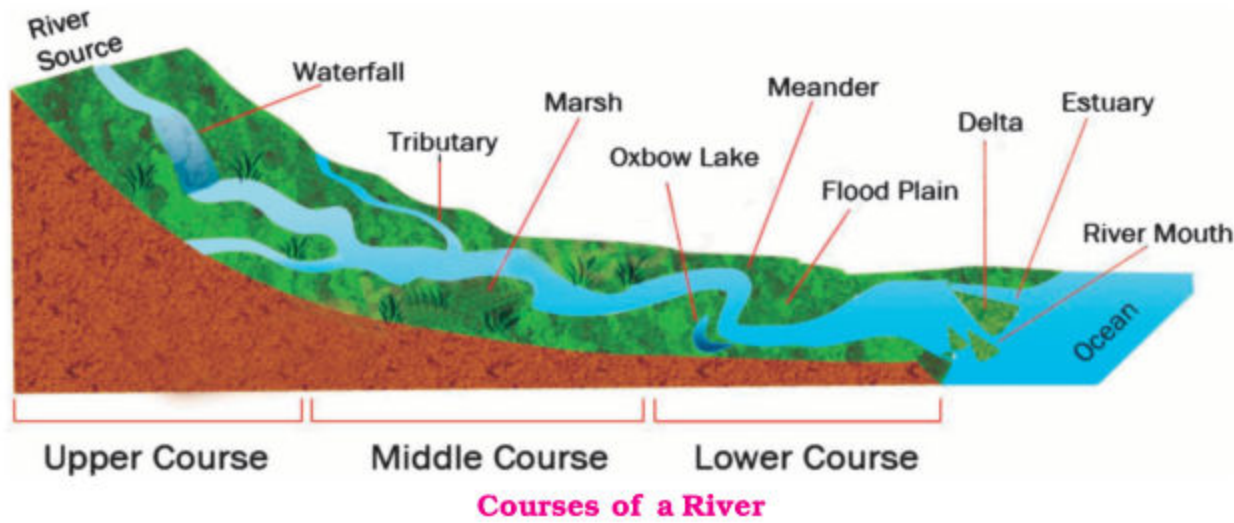
When rock particles carried by water, ice or wind are deposited (dropped) in another location is called deposition.

Why are Rivers Important?

Rivers are extremely important to our society. They provide us with drinking water and irrigation water, help to produce electricity, and allow us to transport material and food by water. These are some of the reasons that major cities and civilizations in the world were established next to a major river. Indus Valley Civilization near River Indus and Cairo on the Nile are the examples.

Courses of Rivers

Rivers are flowing water bodies and found in every continent (except Antarctica). Most of the rivers begin/origin on the hills or mountains. Rivers flow down to the sea and end there. Basically, a river has three parts that are called as courses of a river.



Upper Course	Middle Course	Lower Course
The upper course of a river is often in steep mountainous areas. This section of a river is cold, clear/clean and fast-flowing. When the river flows down through the upper course, the river channel becomes narrow. This creates vertical erosion forming a V-shaped valley. Waterfalls are also usually formed at this stage.	Rivers often flow in meander (follow a winding path) along with their middle course. It is less steep and has a moderate flow of water. Here, the transportation of carried away particles continues and the bottom becomes a mixture of silt and mud known as deposition. At this stage, small meanders or bends and small flood plains can be formed.	It is a course where the river flows gently because of its gentle slope, and it looks like a lake habitat. This winding lowland part of a river contains muddy and slightly warmer water, which flows more slowly. Plants grow in the bed of the river and at its edges. Mainly, deposition takes place here. Larger meanders and larger flood plains as well as oxbow lakes can be formed at this course/stage.

Landscapes Formed by Rivers

There are following landscape features specifically formed by rivers:

V-shaped Valleys: Rivers flow through V-shaped valleys in their upper course. V-shaped valleys are usually found in the mountains and hills. They often have very steep sides. They are formed by erosion. The river carries stones and rocks in its water. The force of the water and the grinding of rocks and stones cut down into the river bed to carve out a valley. Over the time, the valley becomes deeper and wider.



Skardu Valley formed by River Indus

Meanders: Meanders are the curves or bends of a river. They are formed over the time when a river flows around obstructions across a wide valley or flat plain, usually in the middle section of the river.

Oxbow Lake: It is formed when a meander loop gets cut off from the rest of a river. It creates a crescent shaped lake to the side of the river.

Braided Channels: These are a network of small channels. They are formed when the material is deposited from the river causing thinner channels of the river to divert around the deposited material (known as braid bars). These are rarely permanent features and they are frequently altered during flood season.

Flood Plains: An area along the sides of a river which, over time, has been covered with sediments deposited from the river. The sediment which is deposited is called alluvium. During flood season, the flood plain, which tends to be very flat, is likely to be flooded. Adjacent areas of River Indus is its example specially areas of Sindh and Punjab.

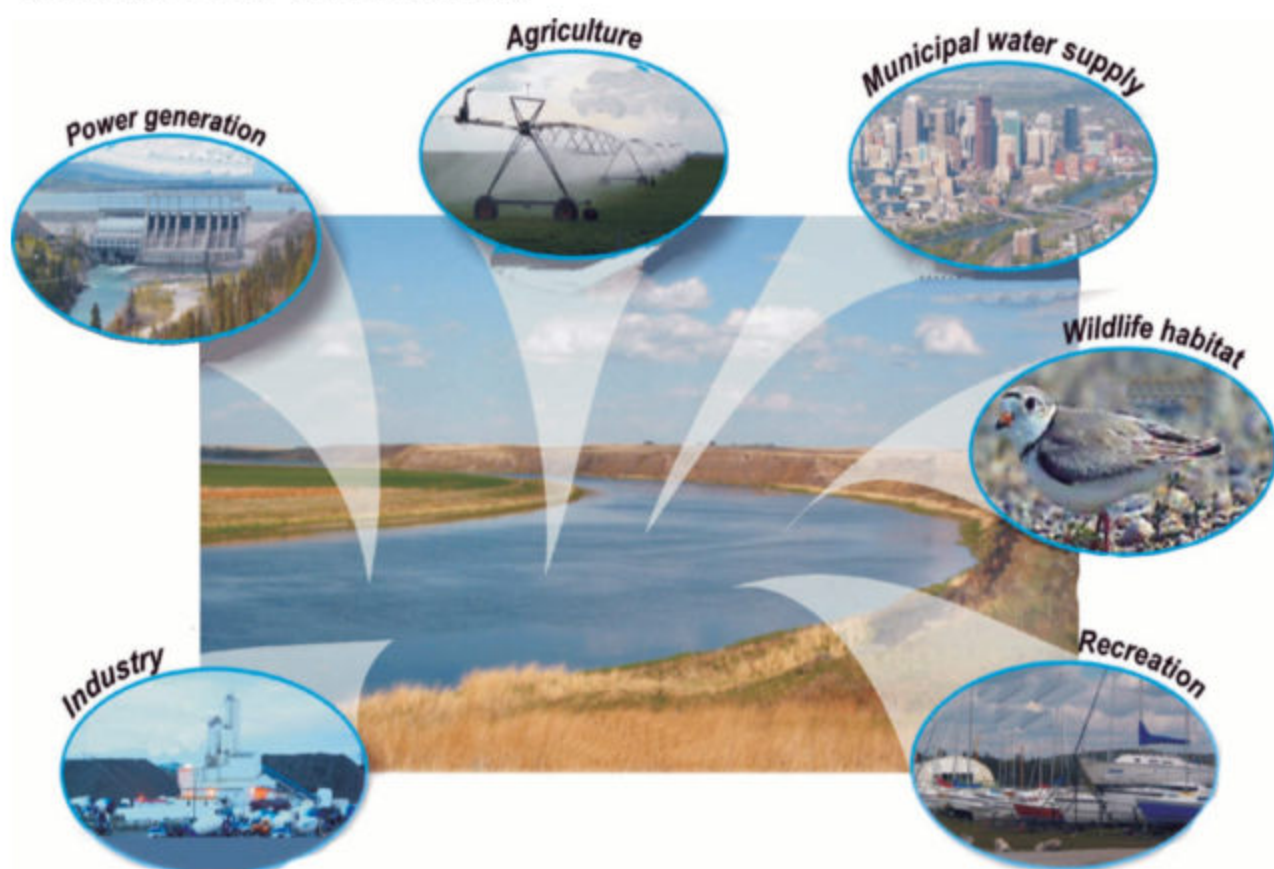
Delta: Look at the figure (Courses of a River) at previous page. It is an area of low flat land, an extension of the flood plain, and often shaped like a triangle where a river splits into smaller rivers as it enters a loch or sea. Deltas are formed when the tides have less impact on the river. Well known examples of deltas include the River Ganges and the Indus Delta.

Estuary: It is where the river mouth widens out and meets the sea. Freshwater from the river and salty sea water are mixed in an estuary. The sea's tidal currents remove much of the sediment from the river mouth and transfer it to the sea. Examples of Estuary Rivers in Scotland include the Rivers Tay, Forth and Clyde.

Alluvial Fan: A large and low, fan shaped deposit of material deposited by a stream, when it flows from a hill or mountain onto a flatter plain. They are found in Northern parts of Pakistan.

How People Use River?

You have studied how important rivers have been to human lives. Rivers have been important in helping people to travel across Earth, especially before modern roads were constructed. People have been producing the energy from rivers. Nowadays, many people use rivers for the purpose of recreation and many have formed groups to help to keep the rivers clean and protect them from damaging industrial pollution. Rivers give us life and help us to look after mankind. It is important that we reciprocate with sincerity.



Activity

Enlist the uses of rivers and explain their effects on human lives.

Teacher's Note

Teacher will hold a discussion on the uses of rivers with students. Then, the teachers will ask students to find out from any available resources individually about the uses of rivers and their effects on human lives. Later, the students will share their findings with the classmates through presentations

Human Activities Affecting River

Damming: Dam is a great renewable source of power generation, but it may destroy river ecosystems downstream. Damming can also be an effective flood control measure to ensure that towns and neighborhoods are safe. Simultaneously, it can obstruct the path of migratory fish and disrupt the life cycle of many different areas.



Dam

Channelising: It is a process to modify a stream or river into a pattern other than what may be its natural meandering. It has a major effect on floodplains.



Channels of a river

Dumping: Storm drains flow directly into the river. Many people don't realize that it makes water "unclean". Discarding materials flow into a river, such as careless disposal of trash and waste pollute rivers. Besides, car oil or antifreeze also pollutes rivers.



Dumping

Domestic Animals: Waste from livestock can be washed into rivers, adding excessive nutrients and illness causing bacteria simultaneously. Pet waste can also be a problem, but they are more likely to impact the wildlife of the river by chasing or hunting.



Water is Beneficial as well as Destructive

Water is one of the most powerful forces of nature. It is a fact that water is our saviour, but water can also be our foe. Water can save lives but it can take lives too. On one hand, it can power underwater turbines that produce electricity. It provides oceans for coral reefs, fish and other sea life. On the other hand, water can be destructive in the form of hurricanes and tsunamis. Sometimes, there is no much time for prevention and precautionary steps. It damages to life and property.

Water in the form of heavy rain can cause floods. People get trapped into floods. Floods have the most devastating effects. Around 180 million people in Pakistan and China were affected by floods in 2010.

How Floods Occur?

A flood occurs when water overflows or inundates the land. Flooding occurs most commonly from heavy rainfall when natural water courses do not have the capacity to convey/transport excess water. However, floods are not always caused by heavy rainfall rather, it can also occur when a river's discharge exceeds the capacity of its channel to carry heavy flow of water from melting glaciers.

Causes of Floods

Flooding may be caused by a number of factors such as:

Increased Urbanization	Bad Farming Practice	Deforestation
		
Increased urbanization has reduced the ability of land to absorb rainfall through the introduction of hard impermeable/ non porous surfaces. This results in increase in the volume and rate of surface run-off as the less water infiltrates in the ground.	Some farmers have caused soil and water to flush into rivers by leaving fields bare and compacting land with cattle and heavy machinery. Even, choosing the wrong direction to plough can cause flooding.	Deforestation (cutting of trees) is responsible for loss of vegetation and soil. The vegetation holds down/intact the soil which acts as a sponge and absorbs more water when it rains.